

# Constitutional AI — A Practical Framework

From values → tests → auditable evidence (without killing velocity)

September 24, 2025

**William Parris**

Keywords: Constitutional AI, Governance, Responsible AI, ModelOps  
Contact: [billparris@gmail.com](mailto:billparris@gmail.com) • [linkedin.com/in/billparris92120](https://www.linkedin.com/in/billparris92120)

# Framework Overview

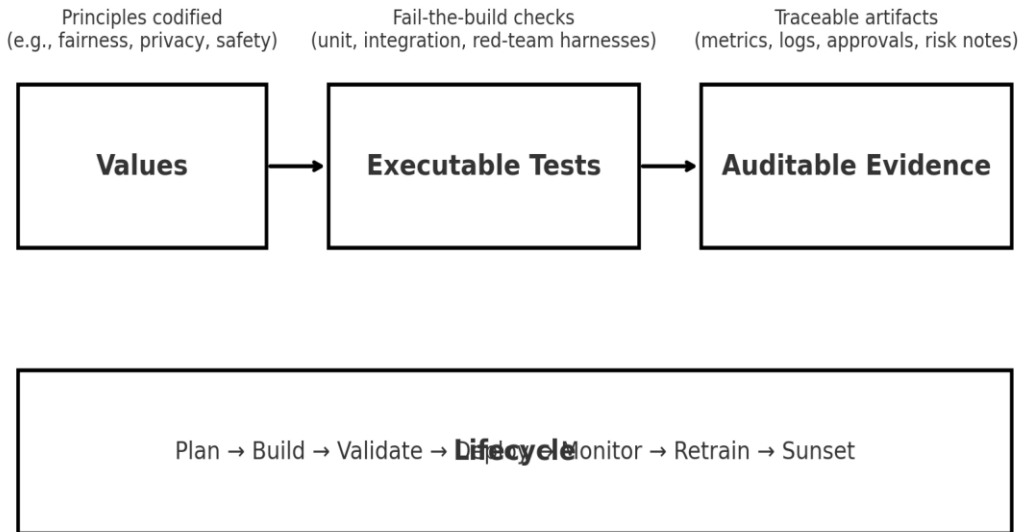


Diagram showing values translating into executable tests and auditable evidence across the model lifecycle (Plan → Build → Validate → Deploy → Monitor → Retrain → Sunset).

## **Core Idea: Values → Tests → Evidence**

The point of Constitutional AI is accountability, not vibes. A usable framework starts by naming values precisely (e.g., privacy, safety, fairness), binds them to executable checks that can fail a build, and leaves an auditable evidence trail that enables reviews, incident response, and learning.

Each value gets a short, testable definition, a handful of acceptance criteria, and a trace to one or more automated or semi-automated checks (unit, integration, adversarial/red-team harnesses). Evidence from these checks is stored with immutable links to model version, data version, approvals, and risk notes.

# Implementation in Five Moves

- 1) Draft a short constitution: 5-9 values with definitions, acceptance criteria, and examples of pass/fail.
- 2) Wire fail-the-build checks: unit guards, integration checks, policy-as-code, and a red-team harness.
- 3) Capture evidence: store metrics, logs, approvals, prompts, and rollbacks with model/data versions.
- 4) Align incentives: make passing governance checks part of the Definition of Done and team KPIs.
- 5) Keep velocity: stage checks to the risk profile; run lightweight guards in dev, expand before release.

# Lifecycle & Roles

Lifecycle checkpoints: Plan (risk register, value selection), Build (policy-as-code, data contracts), Validate (red-team, eval suites), Deploy (approvals, rollback plan), Monitor (drift, incidents), Retrain (changelog, evidence refresh), Sunset (archive, learnings).

Roles (RACI style): Product (accountable for value tradeoffs), ML Eng (responsible for tests), Risk/Safety (advisory + spot checks), Legal/Compliance (constraints), Exec Sponsor (consequence design).

# Metrics & Evidence You Can Trust

Quantitative: task performance deltas, subgroup fairness diffs, privacy leakage scores, abuse precision/recall, latency/throughput, cost-per-K, incident MTTR.

Qualitative: risk notes, red-team excerpts, reviewer rubrics, user feedback samples with stratified sampling.

Evidence gets a stable URI per release (model, data, code, approvals) to make audits and postmortems fast.

# Quick Checklists (Day-One Usable)

Constitution Draft: ☐ values defined; ☐ pass/fail examples; ☐ acceptance criteria; ☐ owners named.

Test Wiring: ☐ unit guards; ☐ integration; ☐ red-team harness; ☐ policy-as-code; ☐ CI/CD gate.

Evidence Trace: ☐ metrics; ☐ logs; ☐ approvals; ☐ incident notes; ☐ versioned artifacts.

Velocity Guardrails: ☐ risk-tiered checks; ☐ fast path in dev; ☐ blocking gate before release.